IT 24633

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Solution of a)

$x \equiv 1 \pmod 3$, $x \equiv 2 \pmod 5$, $x \equiv 3 \pmod 7$

1. start with first two congruences:

let $x = 3a+1$, subsitute into the second congruence

$3a+1 \equiv 2 \pmod 5$

$\Rightarrow 3a = 1 \pmod 5$

The inverse of 3 module 5 is 2 so

$a \equiv 2 \pmod 5$

$\Rightarrow a = 5b+2$

Then $x = 3(5b+2)+1 = 15b+7$

2. Now use third congruence

$15b+7 \equiv 3 \pmod 7$

Since $15 \equiv 1 \pmod 7$ & $7 \equiv 0 \pmod 7$

simplifies to

$b \equiv 3 \pmod 7 \Rightarrow b = 7c+3$

Then $x = 15(7c+3)+7 = 105c+52$

$x \equiv 52 \pmod{15} \Rightarrow 52$. (Ans)

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(2.) Solution of b:

$x \equiv 5 \pmod{11}$, $x \equiv 14 \pmod{29}$, $x \equiv 15 \pmod{31}$

1. Combine the first two congruence

$x = 11a + 5$, subsititure into the second ingruence,

$11a + 5 \equiv 14 \pmod{29}$

$11a = 9 \pmod{29}$

The inverse of 11 nodule 29

That means we want $11x \equiv 1 \bmod 29$

$\therefore 11 \cdot 8 \equiv 1 \bmod 29$

So the inverse is 8. Mutiply both sides:

$a = 9 \times 8 = 72 \equiv 14 \pmod{29}$

So, the inverse is 8. Mutiply both side:

$a = 9 \times 8 = 72 \equiv 14 \pmod{29}$

$a = 29b + 14$

subsitiute back:

$x = 11(29b + 14) + 5 = 319b + 159$

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So, $x \equiv 159 \pmod{319}$

2. Combine with third congruence

we have,

$319b + 159 \equiv 15 \pmod{31}$

$319 \equiv 9 \pmod{31}$ & $159 \equiv 4 \pmod{31}$

So, $9b + 4 \equiv 15 \pmod{31} \Rightarrow 9b \equiv 11 \pmod{31}$

③ Find the inverse of 9 module 31 :

$9 \times 7 = 63 \equiv 1 \pmod{31}$

So, the inverse is 7. multiply both sides :

$b = 11 \times 7 = 77 \equiv 15 \pmod{31} \equiv b = 31c + 15$

substitute back,

$x = 319(31c + 15) + 159$

$= 9889c + 4944$

$\therefore x = 4944 \pmod{9889} \Rightarrow 4944.$ Ans.

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Solution of c)

$x \equiv 5 \pmod 6,\ x \equiv 4 \pmod{11},\ x \equiv 3 \pmod{17}$

1. Combine the first two congruence

$x = 6a + 5$, substitute into the second congruence

$6a + 5 = 4 \pmod{11}$

$6a \equiv -1 \equiv 10 \pmod{11}$

The inverse of 6 module 11 is 2.

Since $6 \times 2 = 12 \equiv 1 \pmod{11}$:

$a = 2 \times 10 = 20 \equiv 9 \mod (11)$

$\Rightarrow a = 11b + 9$

Then $x = 6(11b+9) + 5 = 66b + 59$

2. Now use the 3rd congruence

$66b + 59 \equiv 3 \pmod{17}$

Reduce mod 17

$66 \equiv 15 \pmod{17}$, $59 \equiv 8 \pmod{17}$

So, $15b + 8 \equiv 3 \pmod{17} \Rightarrow 15b = -5 \equiv 17 \pmod{17}$

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The inverse of 15 mod 17 is 8 (since $15 \times 8 = 120$)

$\therefore 15 \times 8 = 120 \equiv 1 \pmod{17}$:

$b = 8 \times 12 = 96 \equiv 11 \pmod{7}$

$\Rightarrow b = 17c + 11$.

Then $x = 66(17c + 11) + 39$

$= 1122c + 785$

$\therefore x \equiv 785 \pmod{1122}$

$\Rightarrow 785$ Ans.